World Journal of Critical Care Medicine

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World J Crit Care Med 2026 September 9; 15(3): 122632

DOI: 10.5492/wjccm.122632

ISSN 2220-3141 (online)

MINIREVIEWS

Venous dimension of shock: Integrating arterial inflow and venous back-pressure in hemodynamic assessment

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Peer review: Externally peer reviewed.
Peer-review model: Single blind
Peer-review report's classification
Scientific quality: Grade B, Grade C
Novelty: Grade C, Grade D
Creativity or innovation: Grade C, Grade C
Scientific significance: Grade B, Grade C
P-Reviewer: Sabath E, Chief, MD, PhD, Mexico; Xu JY, MD, China
Received: April 24, 2026
Revised: June 12, 2026
Accepted: June 25, 2026
Published online: September 9, 2026
Processing time: 126 Days and 0.1 Hours
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Abstract

Critical care hemodynamics has long been centered on arterial pressure, cardiac output, and systemic vascular resistance. While these variables remain fundamental, organ perfusion is determined not only by arterial inflow but also by the pressure opposing venous outflow at the microcirculatory level. Emerging clinical and physiological evidence indicates that elevated venous pressure may contribute to organ dysfunction even when arterial pressure appears adequate. Effective organ perfusion reflects the balance between arterial driving pressure and venous outflow pressure, further modified by microcirculatory factors such as capillary density, flow heterogeneity, and cellular oxygen utilization, which are usually inferred at the bedside rather than directly measured. Consequently, abnormal tissue perfusion may arise from reduced arterial inflow, elevated venous pressure, microcirculatory dysfunction, or a combination of these mechanisms. In this narrative review, we present the arterial-venous perfusion gradient as a pragmatic clinical synthesis that reintegrates established venous physiology into bedside hemodynamic interpretation. The review examines the physiological determinants of venous return, the consequences of venous congestion for organ function, and the clinical conditions in which elevated venous pressure contri-

butes to organ injury. Particular attention is given to the effects of venous congestion on renal, hepatic, splanchnic, cerebral, and right-heart function, alongside bedside tools for evaluating venous hemodynamics, including point-of-care ultrasound, venous Doppler assessment, and ultrasound-based congestion scoring. We also propose a phenotype-based bedside approach for integrating arterial pressure, forward flow, venous congestion, and tissue perfusion during shock assessment and management. Recognizing discordant hemodynamic states may help clinicians identify patients whose organ dysfunction stems from elevated venous back-pressure or persistent microcirculatory impairment, rather than from impaired arterial inflow alone.

Key Words: Arterial-venous perfusion gradient; Central venous pressure; Hemodynamic monitoring; Hypoperfusion; Mean arterial pressure; Microcirculation; Shock; Venous congestion

Core Tip: Venous congestion is an underrecognized cause of organ hypoperfusion in critical illness. This review highlights how elevated venous pressure can reduce organ perfusion despite apparently preserved arterial hemodynamics. The arterial-venous perfusion gradient is presented as a pragmatic clinical synthesis of established physiology, providing a bedside framework to interpret arterial inflow, venous outflow, and tissue perfusion together during shock assessment and resuscitation.

Citation: Kataria S, Vinjamuri S, Juneja D, Goel S. Venous dimension of shock: Integrating arterial inflow and venous back-pressure in hemodynamic assessment. *World J Crit Care Med* 2026; 15(3): 122632
URL: https://www.wjgnet.com/2220-3141/full/v15/i3/122632.htm
DOI: https://dx.doi.org/10.5492/wjccm.122632

INTRODUCTION

Shock is among the most common and most challenging syndromes in critical care. For decades, hemodynamic assessment and resuscitation have focused on the arterial side of the circulation: Mean arterial pressure (MAP), cardiac output, and systemic vascular resistance[1-3]. These parameters underpin contemporary resuscitation algorithms and guide decisions about fluids, vasopressors, and inotropic support[4].

Effective tissue perfusion, however, depends on arterial inflow and the pressure conditions governing venous outflow from the microcirculation. Blood traverses the capillary network between these two pressure domains; the driving force for perfusion is the gradient between them[5-9]. This principle is well established in organ-specific physiology but less consistently applied at the bedside in critically ill patients.

Clinical evidence increasingly links venous congestion to organ dysfunction. Elevated central venous pressure (CVP) has been associated with acute kidney injury (AKI), impaired organ recovery, and worse outcomes across diverse critical care populations[10-13]. These observations support the concept that tissue hypoperfusion may result not only from reduced arterial pressure or impaired forward flow, but also from elevated venous outflow pressure. This distinction is clinically important in patients whose arterial pressure, cardiac output, and tissue perfusion markers do not align, such as those with septic shock after fluid resuscitation, right ventricular dysfunction, obstructive physiology, positive-pressure ventilation, or intra-abdominal hypertension.

In this review, we examine the venous dimension of shock and present the arterial-venous perfusion gradient (AVPG) as a pragmatic clinical synthesis that reintegrates established venous physiology into bedside hemodynamic interpretation. The aim is not to propose new physiological principles, but to provide a clinically useful framework for interpreting arterial pressure, forward flow, venous congestion, and tissue perfusion together. The review summarizes the physiology of venous return and right ventricular function, introduces AVPG-based hemodynamic phenotypes, examines the organ-specific consequences of venous congestion in the kidney, liver, splanchnic circulation, and brain, and discusses bedside tools including point-of-care ultrasound, venous Doppler assessment, and the venous excess ultrasound (VExUS) score. Finally, we translate these concepts into therapeutic implications for fluid administration, vasoactive support, decongestion, microcirculatory reassessment, and extracorporeal support.

LITERATURE SEARCH AND METHODS

This article is a narrative review. We searched PubMed, EMBASE, the Cochrane Library, and Crossref for English-language publications from January 1950 through May 2026, using combinations of the terms venous return, CVP, venous congestion, AVPG, AKI, cardiorenal syndrome, VExUS, point-of-care ultrasound, right ventricular failure, cerebral perfusion pressure (CPP), intra-abdominal hypertension, microcirculation, de-resuscitation, and extracorporeal membrane oxygenation. Reference lists of retrieved articles and recent society guidelines were screened for additional sources. We prioritised peer-reviewed original studies, physiological studies, prospective and observational clinical studies, meta-analyses, and recent literature published between 2020 and 2026. Foundational physiological studies and landmark earlier clinical studies were included when essential to explain venous return, venous congestion, organ

perfusion, or microcirculatory physiology. Reviews and guidelines were used for context and to identify primary sources. Editorials, case reports, and non-English articles were excluded. References were selected for relevance to the AVPG framework rather than for completeness; this review was not registered as a systematic review.

VENOUS SYSTEM: THE CIRCULATORY RESERVOIR AND REGULATOR OF VENOUS RETURN

The venous circulation contains approximately 60%-70% of the total blood volume, making it the principal volume reservoir in the body and a central regulator of cardiac preload and systemic blood flow[14]. Venous return depends critically on the distribution of blood between stressed and unstressed volumes. Stressed volume is the fraction that generates elastic recoil pressure and contributes to venous return, whereas unstressed volume fills the venous capacitance system without generating pressure for flow. Changes in venous tone alter the relative proportions of stressed and unstressed volume without necessarily changing total blood volume[7,14]. In sepsis or under general anesthesia, venodilation increases venous capacitance and reduces the effective stressed volume, thereby impairing effective circulating volume even when total blood volume is normal or increased[7]. In the Guyton model, venous return equals the pressure gradient between mean systemic filling pressure and right atrial pressure, divided by resistance to venous return[5,6,14]. When right atrial pressure rises, this gradient narrows, limiting venous return and cardiac output. Conversely, increasing stressed volume or venous tone widens the gradient and augments venous return, provided that right atrial pressure and resistance to venous return do not rise in parallel.

Several intensive care unit (ICU) specific factors alter venous return and venous pressure. Positive-pressure ventilation raises intrathoracic pressure and, with it, right atrial pressure, reducing venous return, an effect most pronounced in hypovolemic patients[7,8,14-16]. Other conditions (right ventricular failure, pulmonary hypertension, intra-abdominal hypertension) can similarly raise venous pressures and impair circulatory flow. Thus, the venous system should not be viewed as a passive conduit; changes in venous tone, capacitance, pressure, and resistance directly influence cardiac output and hemodynamic stability[6,7,14].

Right ventricular function is central to this interaction. The right ventricle (RV) lies at the downstream end of systemic venous return, and right atrial pressure represents the systemic venous outflow pressure relevant to the macrocirculatory AVPG. Because the RV is thin-walled and afterload-sensitive, it may decompensate rapidly when pulmonary vascular resistance rises, as in pulmonary embolism, acute respiratory distress syndrome, decompensated pulmonary hypertension, RV infarction, or sepsis. A dilated or failing RV raises right atrial pressure, narrows the gradient for systemic venous return, and may produce a venous-congestion phenotype even when MAP is preserved or pharmacologically supported[17].

RV dilation may also impair left ventricular filling through ventricular interdependence. Septal shift and pericardial constraint can reduce left ventricular (LV) diastolic volume independently of intrinsic LV contractility, producing a combined inflow-congestion phenotype in which forward flow is reduced while systemic venous pressure is elevated[18]. At the bedside, reduced tricuspid annular plane systolic excursion, commonly below 17 mm, and reduced tricuspid annular systolic velocity, commonly below 9.5 cm/second, may suggest impaired RV longitudinal function. These measurements should not be interpreted in isolation, but alongside RV size, RV end-diastolic ratio, septal motion, tricuspid regurgitation severity, estimated pulmonary pressures, LV function, ventilatory settings, and the broader clinical context.

ARTERIAL-VENOUS PERFUSION GRADIENT MODEL OF SHOCK

The pressure gradient between arterial inflow and venous outflow across the microcirculation is a fundamental determinant of tissue perfusion. The AVPG framework applies this established principle to bedside shock assessment by interpreting arterial pressure, forward flow, venous back-pressure, and tissue perfusion together. Within this framework, reduced tissue perfusion can result from impaired upstream arterial delivery, elevated downstream venous pressure, or both[8,10,13]. In critically ill patients, these abnormalities frequently coexist, making MAP alone an incomplete marker of effective organ perfusion.

At the bedside, the AVPG can be approximated as the difference between MAP and CVP. CVP is a practical clinical estimate, but venous back-pressure is a broader physiological construct that also encompasses organ-specific pressures and ultrasound-derived congestion markers[8,10,13]. MAP minus CVP should therefore be interpreted as a macrocirculatory surrogate, not a precise measure of organ-level perfusion, which is shaped by microvascular and cellular factors[11,16].

At the tissue level, perfusion depends on microcirculatory function: Capillary density, red blood cell flux, flow heterogeneity, and the diffusion distance for oxygen[19,20]. Venous congestion impairs microcirculatory function by elevating interstitial pressure and promoting tissue edema, thereby increasing the diffusion distance and compressing capillaries [21,22]. Under certain pathological conditions, microvascular flow becomes pressure-dependent, limited by a critical closing pressure below which small vessels collapse and flow ceases[23-25]. Together, these mechanisms account for the frequently observed dissociation between macrocirculatory targets and actual tissue perfusion in the critically ill[11,26].

The AVPG framework therefore does not introduce new physiological principles. Its value lies in providing a clinically oriented synthesis of established physiology, allowing clinicians to recognize situations in which organ dysfunction may be driven by reduced arterial inflow, elevated venous back-pressure, persistent microcirculatory dysfunction, or a combination of these mechanisms. Figure 1 illustrates this conceptual framework.

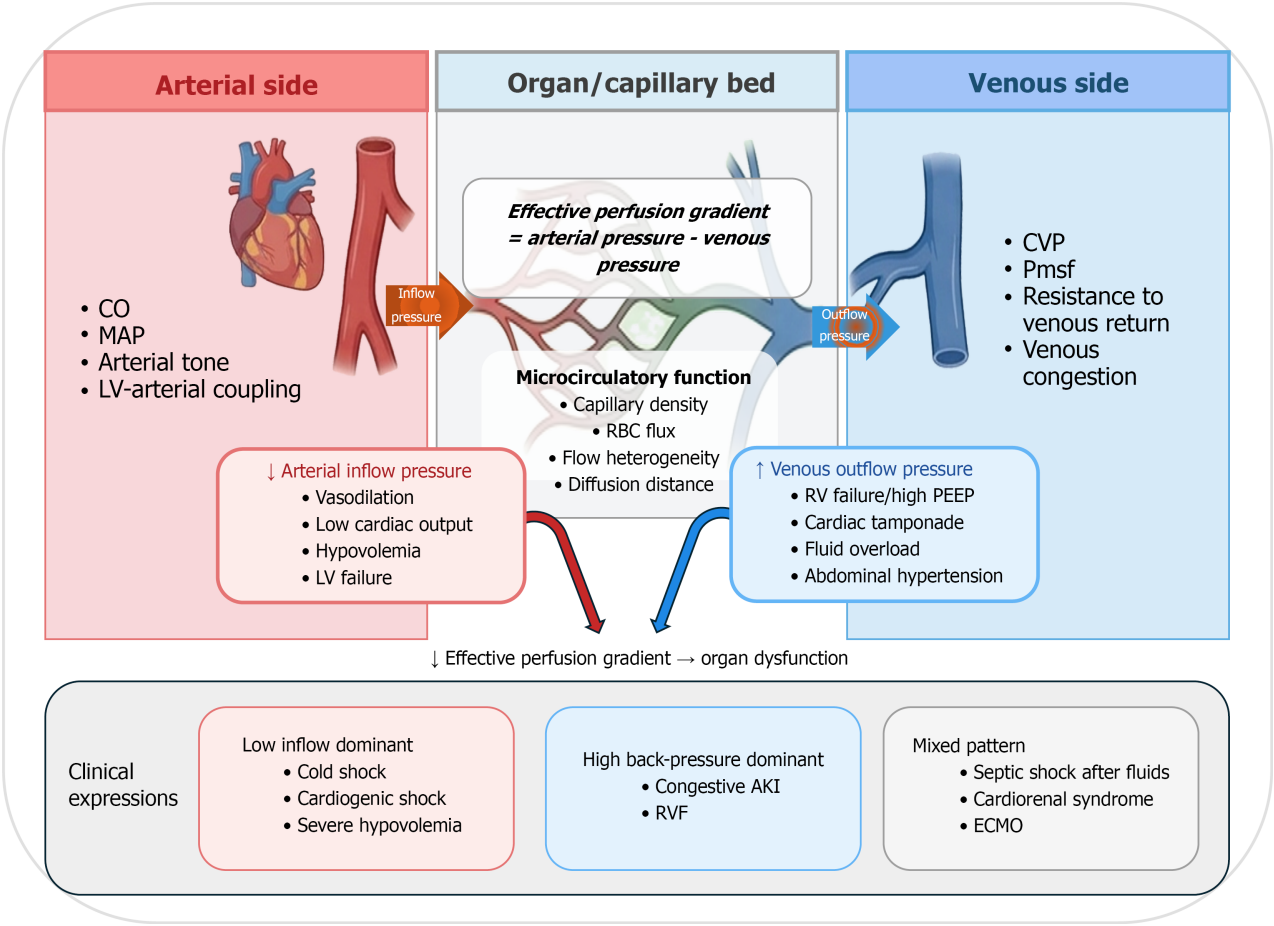


Figure 1 Integrated arterial-venous perfusion gradient model incorporating microcirculatory function. Schematic representation of the relationship between arterial inflow, venous outflow pressure, and the effective perfusion gradient at the level of the organ capillary bed. The gradient is determined by the difference between arterial and venous pressures and is further modulated by microcirculatory factors including capillary density, red blood cell flux, flow heterogeneity, and diffusion distance. Disturbances in arterial inflow, elevations in venous pressure, or their combination reduce the effective perfusion gradient and contribute to organ dysfunction, with distinct clinical expression patterns. AKI: Acute kidney injury; CO: Cardiac output; CVP: Central venous pressure; ECMO: Extracorporeal membrane oxygenation; LV: Left ventricle; MAP: Mean arterial pressure; PEEP: Positive end-expiratory pressure; RVF: Right ventricular failure; RBC: Red blood cell; RV: Right ventricle; Pmsf: Mean systemic filling pressure.

HEMODYNAMIC PHENOTYPES

When arterial inflow and venous outflow pressures are considered together, circulatory failure can be categorized into four hemodynamic patterns based on the relative contributions of each, which together determine the effective organ perfusion gradient (Figure 2).

Normal perfusion

In the normal circulatory state, arterial inflow and venous outflow pressures are balanced such that the effective perfusion gradient across tissues is preserved. Cardiac output and arterial pressure are sufficient to maintain systemic blood flow, while venous pressures remain low enough to permit adequate drainage from the microcirculation. Under these conditions, organ perfusion and oxygen delivery are maintained, and markers of tissue perfusion remain normal.

Reduced arterial inflow

Here, tissue hypoperfusion results from reduced arterial inflow pressure. Cardiac output, circulating volume, or effective arterial pressure fall, reducing the upstream driving force for blood flow. Venous pressures remain relatively normal, so the narrowed perfusion gradient is attributable to the arterial side. Hypovolaemic shock, cardiogenic shock, and early distributive shock all fit this pattern. Anaphylactic and neurogenic shock present similarly when sudden venodilation and loss of sympathetic tone collapse mean systemic filling pressure. Patients typically present with hypotension, low cardiac output, elevated lactate, prolonged capillary refill time (CRT), or altered mental status[1].

Venous congestion

In this pattern, elevated venous pressure narrows the perfusion gradient despite relatively preserved or supportable arterial pressure. The increased downstream pressure opposes drainage from the microcirculation. Any condition causing systemic venous hypertension can produce this state. Obstructive shock from massive pulmonary embolism, cardiac

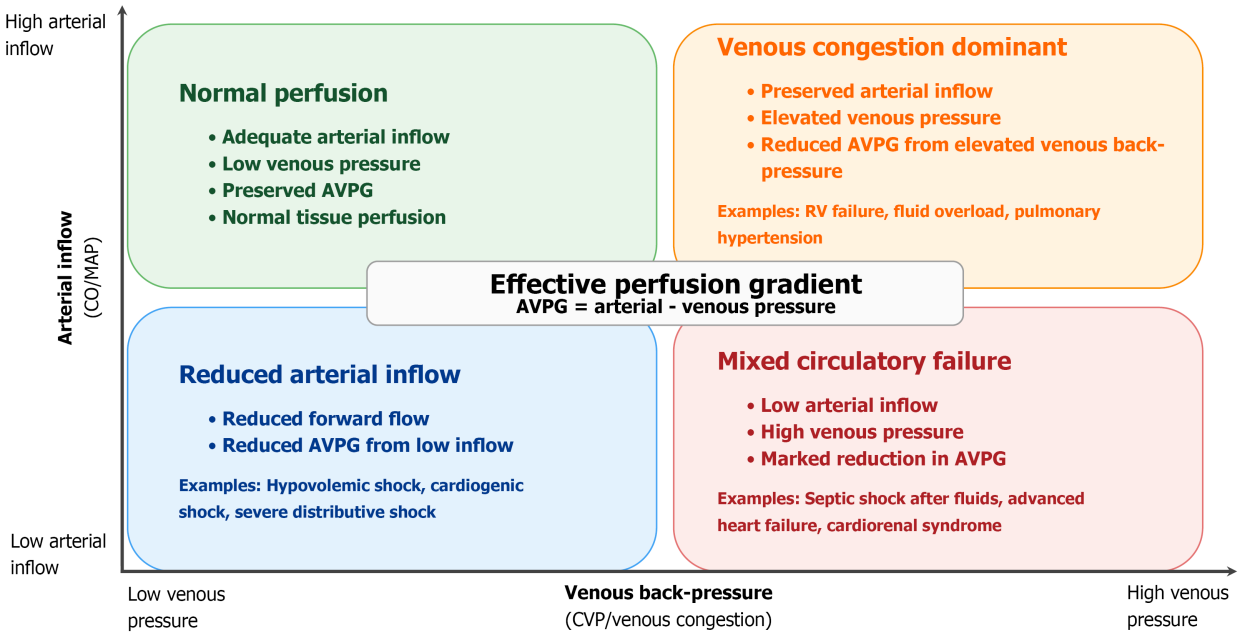


Figure 2 Hemodynamic phenotypes based on the interaction between arterial inflow and venous outflow pressure. Schematic representation of four hemodynamic patterns defined by arterial inflow and venous pressure, which together determine the effective organ perfusion gradient. Impairment in either or both components leads to reduced tissue perfusion. AVPG: Arterial-venous perfusion gradient; CO: Cardiac output; CVP: Central venous pressure; MAP: Mean arterial pressure.

tamponade, or tension pneumothorax and isolated right-sided cardiogenic shock are the prototypical examples; over-resuscitated septic shock produces similar physiology by iatrogenic volume loading. Patients present with elevated CVP, clinical signs of venous congestion, and progressive organ dysfunction, often without marked arterial hypotension[10,27].

Combined inflow-congestion failure

Frequently, both mechanisms are present simultaneously. Cardiac output or arterial pressure is reduced at the same time that venous pressure is elevated, compressing the perfusion gradient from both ends. Advanced heart failure, late septic shock after aggressive fluid resuscitation, and complex cardiopulmonary interactions involving ventricular dysfunction with systemic venous congestion are common clinical scenarios. Obstructive shock crosses into this phenotype when reduced left-sided output from RV failure or tamponade coexists with elevated systemic venous pressures. These patients tend to be hemodynamically unstable and at high risk for progressive organ failure[28-30].

Microcirculatory dysfunction as a downstream modifier

The above phenotypes describe macrocirculatory patterns of arterial inflow, venous outflow, and the resulting organ perfusion gradient. Microcirculatory dysfunction should not be viewed as a separate venous-pressure phenotype, but as a downstream modifier that may complicate any of these patterns. In some patients, tissue hypoperfusion persists despite apparently adequate arterial pressure, forward flow, and venous pressure control. This state reflects loss of hemodynamic coherence, in which correction of macrocirculatory variables does not fully restore capillary perfusion or tissue oxygen utilization. Clinically, this may be suggested by persistent capillary refill delay, mottling, rising or non-clearing lactate, oliguria, altered mentation, or organ dysfunction despite correction of MAP, cardiac output, and venous congestion. In such cases, further fluid loading or pressure escalation should not be automatic; instead, clinicians should reassess oxygen delivery, hemoglobin, hypoxemia, acidosis, temperature, source control, inflammatory injury, excessive vasoconstriction, and metabolic dysfunction.

These phenotypes are not mutually exclusive but rather represent different points along a spectrum of circulatory failure. Identifying the predominant pattern in a given patient can direct therapy more precisely (Table 1).

VENOUS CONGESTION AND ORGAN DYSFUNCTION

As outlined in the AVPG framework above, the main microcirculatory mechanisms by which venous congestion drives organ injury are elevated capillary hydrostatic pressure, interstitial oedema, capillary compression, lengthening of the oxygen diffusion distance, and a shift toward pressure-dependent microvascular flow[13,22,23]. The relative weight of these mechanisms varies by organ and by clinical setting, as detailed below.

Table 1 Hemodynamic phenotypes based on disturbances in arterial inflow and venous pressure

Phenotype	Arterial inflow	Venous pressure	AVPG	Typical clinical settings	Key bedside features
Normal perfusion	Adequate	Normal	Preserved	No clinically relevant mismatch between arterial inflow and venous outflow	Normal perfusion markers, preserved urine output, stable mentation, and no evolving organ dysfunction
Reduced arterial inflow	Decreased	Normal or low	Decreased mainly from the arterial side	Hypovolemia, cardiogenic shock, and distributive shock with impaired effective flow	Hypotension, low cardiac output, elevated or rising lactate, prolonged capillary refill time, oliguria
Predominant venous congestion	Preserved or relatively preserved	Increased	Decreased mainly from the venous side	Right ventricular failure, pulmonary hypertension, obstructive shock, fluid overload, and over-resuscitated septic shock	Elevated or rising CVP, systemic venous congestion, abnormal venous Doppler or VExUS, oliguria, renal or hepatic dysfunction
Combined inflow-congestion failure	Decreased	Increased	Markedly decreased from both sides	Advanced heart failure, late septic shock after fluid loading, RV failure with impaired LV filling, and complex cardiopulmonary failure	Hypotension or low forward flow with systemic venous congestion, elevated CVP, organ dysfunction, and persistent hypoperfusion

Representative evidence supporting these phenotypes is drawn from studies and guidelines on shock classification, venous congestion, right ventricular failure, fluid responsiveness, and septic/cardiogenic shock phenotyping[1-4,9,10,13,17,27-30,34]. These phenotypes represent dominant patterns rather than mutually exclusive categories. Microcirculatory dysfunction may complicate any phenotype and should be suspected when tissue hypoperfusion persists despite apparent correction of arterial pressure, forward flow, and venous congestion. AVPG: Arterial-venous perfusion gradient; CVP: Central venous pressure; RV: Right ventricle; VExUS: Venous excess ultrasound.

Kidney

The kidney is particularly sensitive to elevations in venous pressure. Experimental work has shown that increases in renal venous pressure significantly reduce renal blood flow and glomerular filtration rate even when arterial pressure is preserved[31]. In acute decompensated heart failure, Mullens and colleagues reported that worsening renal function tracked venous congestion more closely than reductions in cardiac output[10], and Damman *et al*[32] confirmed that elevated CVP is associated with renal impairment across a broad cardiovascular spectrum. Observational ICU cohorts and a recent meta-analysis have identified elevated CVP as an independent risk factor for AKI[13], and a renal perfusion pressure concept (MAP minus the greater of CVP or intra-abdominal pressure) is increasingly used as a bedside target in critical illness[33]. Renal venous congestion is particularly relevant in cardiorenal syndrome type 1 and in over-resuscitated septic shock; in hypovolaemic shock, AKI is more typically pre-renal from reduced inflow than from congestion[34].

Liver and splanchnic circulation

Elevated hepatic venous pressure produces sinusoidal congestion and hepatocellular injury, the classic substrate of congestive hepatopathy[35]. In critically ill patients, mesenteric venous hypertension contributes to intestinal oedema, reduces splanchnic microvascular flow, and impairs gut barrier function, facilitating bacterial translocation and systemic inflammation[36]. Normal portal venous flow is continuous and only mildly pulsatile; on point-of-care ultrasound, a portal pulsatility fraction above 30% is considered abnormal and a fraction above 50% denotes severe transmitted right-atrial pressure, although neither is entirely specific[37]. Intra-abdominal hypertension raises hepatic and renal venous pressures simultaneously, contributing to a cardio-abdomino-renal axis re-emphasised in recent critical-care literature [38].

Brain

CPP is MAP minus the greater of intracranial pressure or the downstream cerebral venous outflow pressure. Any systemic event that raises ICP through impaired cerebral venous outflow narrows CPP at constant MAP. Elevated CVP, jugular venous hypertension from head-down positioning or rigid cervical collars, and intra-abdominal hypertension can each impede cranial venous drainage and raise ICP[39]. High positive end-expiratory pressure and prone positioning during acute respiratory distress syndrome or after cardiac arrest can elevate intrathoracic pressure enough to impair jugular outflow, with measurable CPP consequences in patients with reduced cerebral compliance[40].

Lungs and right heart interactions

Several ICU conditions converge to elevate CVP: Right ventricular dysfunction, pulmonary hypertension, positive-pressure ventilation, intra-abdominal hypertension, and excessive fluid administration[41-43]. The RV discussion above details the mechanism. Importantly, venous congestion often develops progressively during critical illness; patients initially resuscitated with aggressive fluid therapy may subsequently develop elevated venous pressures that compromise organ perfusion despite apparently adequate arterial pressure and cardiac output.

INTEGRATED BEDSIDE HEMODYNAMIC ASSESSMENT

Bedside evaluation of circulatory status in critically ill patients relies on the integration of clinical examination, hemodynamic monitoring, and point-of-care imaging, as reflected in recent European Society of Intensive Care Medicine guidelines on circulatory shock[44]. These complementary approaches provide information on arterial pressure, cardiac output, and venous pressure, allowing clinicians to better understand the mechanisms underlying circulatory instability.

Assessment of arterial hemodynamics traditionally forms the foundation of shock evaluation. Measurement of MAP provides an estimate of upstream perfusion pressure, while cardiac output monitoring, using techniques such as echocardiography, pulse contour analysis, or thermodilution, offers insight into forward blood flow and global oxygen delivery [1,2].

Additional bedside indicators may provide indirect markers of tissue perfusion and help identify states of impaired circulatory flow. Serum lactate levels are widely used as a marker of global tissue hypoxia or impaired oxygen utilization, while CRT and peripheral perfusion indices provide simple clinical indicators of microvascular perfusion[45,46]. The veno-arterial carbon dioxide tension difference (Pv-aCO_2 gap) may offer additional insight into the adequacy of circulatory flow. An increased CO_2 gap reflects impaired tissue clearance of CO_2. It has been associated with states of low cardiac output or inadequate blood flow relative to metabolic demand[47,48]. When interpreted alongside markers of oxygen metabolism such as lactate and central venous oxygen saturation, the CO_2 gap may help identify patients in whom forward flow remains insufficient despite apparently adequate arterial pressure[47]. From a physiological perspective, an increased CO_2 gap reflects inadequate circulatory flow relative to metabolic demand. It may therefore signal conditions in which the effective pressure gradient driving tissue perfusion is insufficient to sustain adequate blood flow.

Direct bedside assessment of the microcirculation is not yet a routine part of clinical care. Hand-held videomicroscopy (sidestream dark-field and incident dark-field imaging) quantifies perfused capillary density, the proportion of perfused vessels, and a heterogeneity index, and has shown that microcirculatory alterations are common in septic shock and predict organ failure[9,49,50]. These methods remain largely research tools; a recent multicentre randomised trial of direct microcirculation-guided shock resuscitation showed no clear outcome advantage over usual care[51]. Bedside surrogates (CRT, peripheral perfusion index, serum lactate, and the Pv-aCO_2 gap) have important limitations of their own, although CRT-guided resuscitation tested favourably in ANDROMEDA-SHOCK and was further evaluated in ANDROMEDA-SHOCK-2[45,47]. These limitations underlie the dissociation between corrected macrohemodynamics and ongoing tissue hypoperfusion described as hemodynamic incoherence[9,26].

Point-of-care ultrasound in hemodynamic phenotyping

Cardiac point-of-care ultrasound allows rapid bedside estimation of LV systolic function, LV outflow tract velocity-time integral as a surrogate for stroke volume, RV size and function (including the RV:LV ratio and septal motion), and pericardial effusion or tamponade. Lung ultrasound adds B-lines as a marker of extravascular lung water and pulmonary venous congestion, A-line predominance in dry lungs, and pleural effusion. A recent meta-analysis confirmed that integrated point-of-care ultrasound distinguishes shock subtypes with good diagnostic accuracy[52]. Echocardiography is also the practical means of detecting and following the right-ventricular phenotypes that drive the venous side of the AVPG.

Evaluation of the venous circulation provides complementary information regarding downstream pressures that may influence organ perfusion. CVP, measured through a central venous catheter, reflects right atrial pressure and may indicate systemic venous hypertension when elevated. Although CVP has limitations as a marker of intravascular volume status, it may provide useful information when interpreted in conjunction with arterial pressure, cardiac output, and the broader clinical context[48].

Point-of-care ultrasound has emerged as an important tool for assessing systemic venous congestion. Ultrasound examination of the inferior vena cava can provide information regarding venous filling and right atrial pressure. Although this interpretation may be influenced by factors such as mechanical ventilation and changes in intrathoracic pressure[53]. Doppler interrogation of abdominal veins, including the hepatic, portal, and intrarenal veins, can provide additional insights into the transmission of elevated venous pressure to organ circulation.

VExUS

The VExUS score grades systemic venous congestion by combining an inferior vena cava diameter of at least 2 cm with Doppler abnormalities in the hepatic, portal, and intrarenal veins[37,54]. Grades run from 0 (no congestion) to 3 (severe, with two or more severely abnormal Doppler patterns). Prospective work has shown that VExUS correlates with invasive right atrial and pulmonary capillary wedge pressures[55], and higher grades are associated with AKI in cardiac surgery and general ICU cohorts[56,57]. Limitations are real: Isolated tricuspid regurgitation, atrial fibrillation, cirrhosis, and abdominal compartment syndrome each confound one or more components, intrarenal vein imaging is the most operator-dependent step, and no randomised trial has yet shown that VExUS-guided decongestion improves patient-centred outcomes. Serial VExUS during de-resuscitation may help track changes in congestion and inform diuretic titration.

Table 2 summarizes bedside variables and pragmatic reassessment triggers for integrated AVPG-based assessment. These values should not be treated as rigid diagnostic or therapeutic thresholds. Rather, they are intended to prompt reassessment of the dominant hemodynamic phenotype and to guide whether further fluid administration, vasoactive support, right ventricular-directed therapy, venous-pressure reduction, or microcirculatory reassessment is most appropriate.

Table 2 Bedside variables and pragmatic reassessment triggers for integrated arterial-venous perfusion gradient-based hemodynamic assessment

Domain	Bedside variable/tool	Pragmatic reassessment trigger	Clinical implication
Arterial pressure	MAP, arterial waveform quality, vasopressor requirement	MAP < 65 mmHg, poor arterial waveform, low diastolic pressure, or escalating vasopressor dose	Confirm measurement accuracy and interpret MAP with tissue perfusion, baseline blood pressure, cardiac output, CVP, and shock phenotype. MAP alone does not confirm adequate organ perfusion
Forward flow	Cardiac output, LVOT-VTI, focused echocardiography, pulse pressure, $ScvO_2$	Low cardiac output or low LVOT-VTI for clinical context; low $ScvO_2$ with persistent hypoperfusion	Suggests inadequate forward flow. Reassess preload, contractility, obstruction, RV function, hemoglobin, oxygenation, and metabolic demand
Fluid responsiveness	Passive leg raise or small fluid challenge with SV/CO/LVOT-VTI measurement	SV/CO/LVOT-VTI increase of approximately 10%-15%	Suggests that additional preload can increase forward flow. A fluid bolus should be repeated only if the patient is also fluid tolerant.
Fluid tolerance	CVP trend, MAP-CVP, VExUS, RV size/function, lung ultrasound, oxygenation, fluid balance	Rising CVP, falling MAP-CVP, VExUS grade 2-3, worsening RV dilation/septal shift, new or increasing bilateral B-lines, worsening oxygenation, or large positive fluid balance	Suggests limited tolerance to further fluid. Shift from "Is the patient fluid responsive?" to "Can the patient tolerate more fluid?"
Tissue perfusion	Lactate, CRT, mottling, skin temperature, urine output, mental status	Lactate ≥ 2 mmol/L, rising or non-clearing lactate, CRT > 3 seconds, mottling, urine output < 0.5 mL/kg/hour, altered mentation, or worsening organ function	Suggests persistent hypoperfusion or metabolic stress. Reassess arterial pressure, forward flow, venous congestion, and microcirculatory coherence rather than giving fluid automatically
Flow-metabolism relationship	Pv-aCO_2 gap, $ScvO_2$, lactate trend	Pv-aCO_2 gap > 6 mmHg, especially with raised lactate or low $ScvO_2$	May suggest inadequate blood flow relative to metabolic demand. Reassess cardiac output, oxygen delivery, hemoglobin, hypoxemia, and source control
Venous pressure/AVPG surrogate	CVP and MAP-CVP trend	CVP > 12-15 mmHg or rising; low or falling MAP-CVP, especially < 50 mmHg with organ dysfunction	Suggests venous back-pressure may be contributing to organ dysfunction. Assess RV function, VExUS, intra-abdominal pressure, ventilator pressures, and fluid balance
Venous congestion ultrasound	IVC, hepatic, portal, and intrarenal venous Doppler; VExUS score	Dilated IVC with abnormal venous Doppler, portal pulsatility > 30%, discontinuous or monophasic intrarenal venous flow, or VExUS grade 2-3	Supports clinically relevant venous congestion. Avoid unnecessary fluid loading and consider venous-pressure reduction when perfusion pressure is supportable
Right ventricular assessment	Focused echocardiography: RV size/function, RV:LV ratio, septal motion, TAPSE, S', TR	RV dilation, RV:LV ratio > 1, septal flattening/shift, TAPSE < 17 mm, S' < 9.5 cm/seconds, significant TR, or pulmonary pressure overload	Suggests RV-mediated venous hypertension or combined inflow-congestion failure. Avoid blind fluid loading; optimize RV preload, afterload, contractility, and systemic pressure
Extrinsic contributors to venous pressure	PEEP, auto-PEEP, dynamic hyperinflation, intra-abdominal pressure, tamponade or tension physiology	High PEEP or auto-PEEP, intra-abdominal pressure ≥ 12 mmHg, tamponade physiology, tension pneumothorax, or abdominal compartment physiology	These factors can raise venous pressure and impair organ drainage despite acceptable MAP. Correcting the driver may improve the effective AVPG
Pulmonary fluid tolerance	Lung ultrasound B-lines, oxygenation, ventilatory requirement, and EVLW where transpulmonary thermodilution is available	New or increasing bilateral B-lines after fluid, worsening oxygenation, or rising EVLW or increasing ventilatory support	Suggests reduced pulmonary fluid tolerance. B-lines should not be interpreted as systemic venous congestion alone but should prompt reassessment before further fluid administration. Where transpulmonary thermodilution is available, extravascular lung water may provide additional quantitative information on pulmonary fluid accumulation and fluid tolerance, but it should be interpreted as an adjunct rather than a direct measure of systemic venous congestion
Residual microcirculatory dysfunction	Persistent CRT delay, mottling, lactate non-clearance, oliguria, altered mentation, or organ dysfunction despite corrected macrocirculation	Persistent hypoperfusion despite MAP ≥ 65 mmHg, adequate or improved forward flow, and no major venous congestion	Suggests possible loss of hemodynamic coherence. Reassess source control, oxygen delivery, hemoglobin, hypoxemia, acidosis, temperature, excessive vasoconstriction, and inflammatory or metabolic injury

Suggested reassessment triggers are based on guideline recommendations, physiological studies, observational data, and ultrasound literature; they are intended as pragmatic triggers rather than validated universal treatment thresholds[4,12,34,37,44-59]. They should be interpreted with shock etiology, baseline blood pressure, right ventricular function, venous Doppler findings, tissue perfusion, and response to intervention. AVPG: Arterial-venous

perfusion gradient; CO: Cardiac output; CRT: Capillary refill time; CVP: Central venous pressure; IAP: Intra-abdominal pressure; IVC: Inferior vena cava; LVOT-VTI: Left ventricular outflow tract velocity-time integral; EVLW: Extravascular lung water; MAP: Mean arterial pressure; PEEP: Positive end-expiratory pressure; PLR: Passive leg raise; Pv-aCO_2: Veno-arterial carbon dioxide tension difference; RV: Right ventricle; $ScvO_2$: Central venous oxygen saturation; SV: Stroke volume; TAPSE: Tricuspid annular plane systolic excursion; TR: Tricuspid regurgitation; VExUS: Venous excess ultrasound.

THERAPEUTIC IMPLICATIONS: TARGETING THE ARTERIAL-VENOUS PERFUSION GRADIENT

The AVPG framework allows therapy to be matched to the dominant physiological mechanism. Bedside hemodynamic evaluation should therefore address three domains simultaneously: Arterial inflow (MAP and cardiac output), venous outflow pressure (congestion indicators), and tissue reperfusion markers (CRT, lactate kinetics, venous-to-arterial CO_2 difference)[46,47,58]. The aim is not to introduce a new numerical resuscitation target, but to guide whether the next intervention should be further fluid administration, vasoactive support, venous-pressure reduction, right ventricular-directed therapy, or reassessment for persistent microcirculatory dysfunction.

Further fluid administration should be considered only when two conditions are met: First, additional preload is likely to increase forward flow; and second, the patient remains fluid tolerant. A passive leg raise or small fluid challenge should therefore be coupled with stroke volume, cardiac output, pulse pressure, or left ventricular outflow tract velocity-time integral (LVOT-VTI) assessment. A rise in stroke volume, cardiac output, or LVOT-VTI of approximately 10%-15% supports fluid responsiveness, but this finding should not be interpreted in isolation[59]. If CVP is > 12-15 mmHg or rising, the MAP-CVP gradient is falling, VExUS suggests significant congestion, right ventricular dilation is present, oxygenation is worsening, or new/increasing bilateral B-lines appear after fluid administration, repeated boluses may be avoided even if arterial pressure remains low.

The practical transition from resuscitation to decongestion should be based on concordant findings rather than on a single CVP cutoff. Further fluid becomes progressively less appropriate when absent or weak fluid responsiveness coexists with rising venous pressure, low or falling MAP-CVP, VExUS grade 2-3, hepatic, portal, or intrarenal venous Doppler congestion, positive fluid balance, pulmonary fluid intolerance, or worsening renal, hepatic, or tissue perfusion markers despite arterial pressure support[13,28,34,54-57,60-63]. At this point, the bedside question should shift from "Does this patient need more fluid?" to "Is venous pressure now limiting organ perfusion?"

Combined inflow-congestion failure requires a balanced approach because interventions may improve one side of the gradient while worsening the other. Excessive fluid may transiently increase arterial pressure but worsen venous congestion and organ perfusion, whereas overly aggressive decongestion may reduce preload and compromise forward flow[28,62]. In this phenotype, hypotension should not be treated with fluid alone. Arterial pressure should be supported, usually with norepinephrine, while cardiac output, right ventricular function, obstruction, ventilatory pressures, intra-abdominal pressure, and venous congestion are reassessed in parallel. When norepinephrine requirements are high or escalating, particularly at doses of approximately > 0.3-0.5 μg/kg/minute, and the patient has rising CVP, VExUS grade 2-3, worsening renal or hepatic function, or persistent hypoperfusion, vasoplegia and congestion may coexist. In this setting, adding vasopressin may help limit catecholamine escalation, while cautious decongestion can be pursued once perfusion pressure is supportable[4,64,65].

Right ventricular dysfunction should be treated as a central mechanism of venous congestion rather than as a secondary observation. A dilated RV, RV/LV ratio > 1, septal shift, severe tricuspid regurgitation, pulmonary hypertension, high CVP, or abnormal hepatic, portal, or intrarenal venous Doppler should prompt avoidance of blind fluid loading. Management should focus on optimizing rather than maximizing preload, maintaining systemic arterial pressure for right ventricular coronary perfusion, correcting hypoxemia, hypercapnia, and acidosis, reducing excessive positive end-expiratory pressure or dynamic hyperinflation where feasible, and considering inotropes or pulmonary vasodilators when impaired right ventricular contractility or elevated pulmonary vascular resistance is dominant[16,17,30,40,41]. Obstructive causes such as massive pulmonary embolism, tamponade, or tension pneumothorax require definitive relief of obstruction, with vasoactive therapy used as a bridge[66].

AVPG-guided priorities differ by shock type. Hypovolemic shock primarily requires restoration of arterial inflow, with venous congestion becoming relevant mainly after excessive or ongoing fluid administration. Cardiogenic shock often requires simultaneous forward-flow support and decongestion[67,68]. Septic shock may evolve over time, from early vasopressor-supported volume resuscitation to later de-resuscitation if fluid accumulation and venous congestion develop[62,63]. Obstructive shock requires definitive correction of the obstructive process, whereas anaphylactic and neurogenic shock require restoration of vascular tone and effective stressed volume. This phenotype-based distinction is important because venous congestion should not be managed identically across all shock states[1-4,62,63,67,69,70].

Extracorporeal support modifies venous and arterial components of the AVPG. Veno-arterial extracorporeal membrane oxygenation may reduce systemic venous pressure when venous drainage is adequate, but retrograde arterial reinfusion can increase left ventricular afterload and worsen pulmonary venous congestion if the left ventricle is not unloaded[69,71]. Persistent venous congestion during VA-ECMO should prompt reassessment of cannula position, circuit drainage, right and left ventricular function, intra-abdominal pressure, ventilator settings, and fluid balance. Veno-venous ECMO does not directly unload the systemic venous circulation, although improved oxygenation may permit reduction in airway pressures and indirectly improve venous return.

Finally, microcirculation should be used as the downstream check of whether correction of pressure, flow, and venous congestion has translated into tissue perfusion. If MAP, forward flow, MAP-CVP, and venous congestion have improved but CRT, mottling, lactate, urine output, mental status, or organ function remain abnormal, persistent microcirculatory dysfunction or loss of hemodynamic coherence should be considered[14,26,45,47,58]. In this situation, the response

should not be automatic fluid loading or further vasopressor escalation. The clinician should reassess source control, oxygen delivery, hemoglobin, oxygenation, acid-base status, temperature, metabolic stress, excessive vasoconstriction, and ongoing inflammatory injury.

Thus, AVPG-based management moves bedside decision-making from a pressure-only approach to a perfusion-gradient approach. Once the dominant phenotype has been recognized, therapy should be directed toward cautious fluid administration, arterial pressure support, forward-flow optimization, venous-pressure reduction, right ventricular-directed therapy, correction of obstruction or extrinsic pressure, or reassessment for residual microcirculatory dysfunction. This sequence is summarized in Figures 3 and 4.

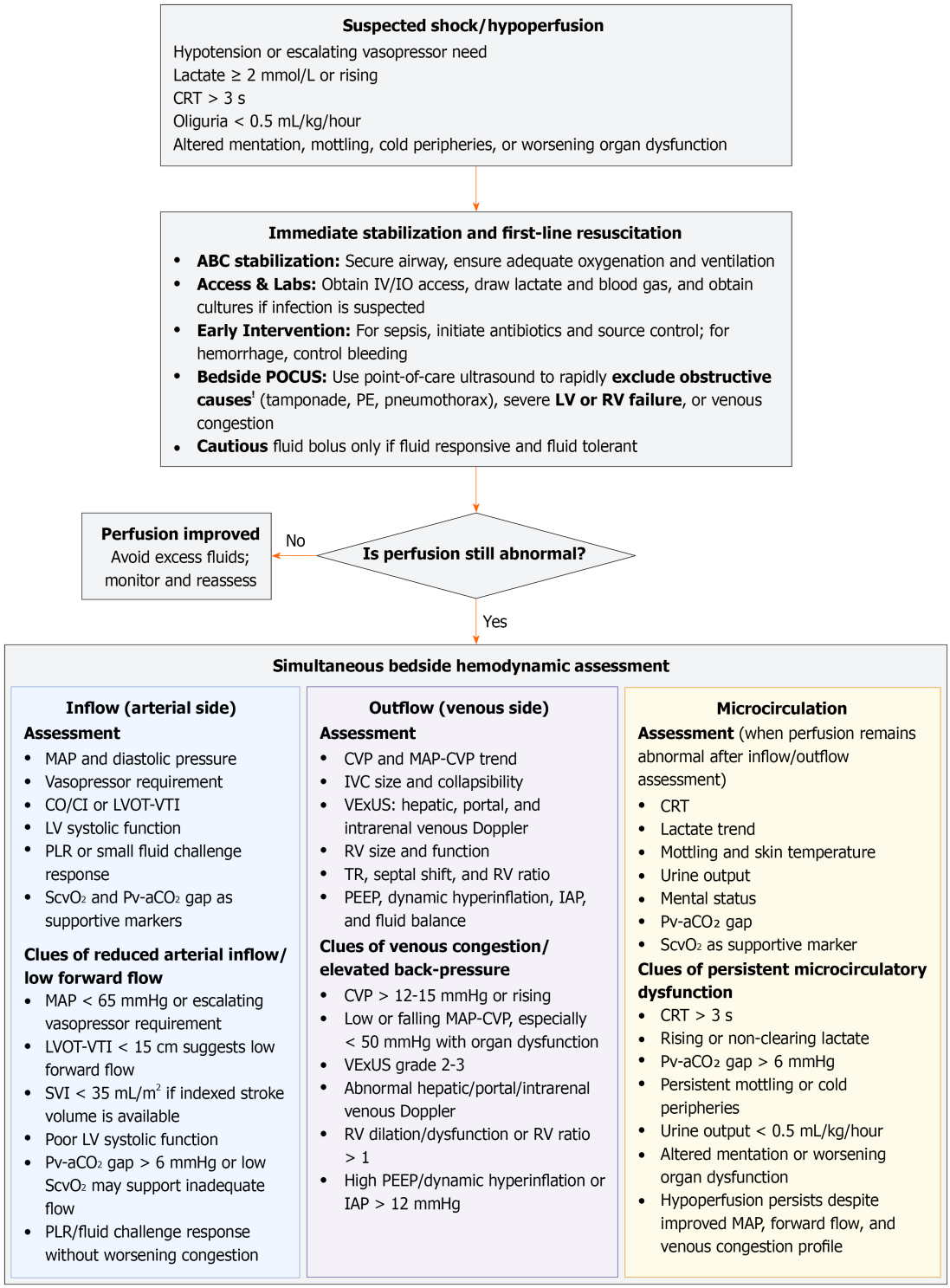


Figure 3 Simultaneous arterial-venous perfusion gradient-based bedside assessment of suspected shock or tissue hypoperfusion. After initial stabilization, persistent hypoperfusion should be assessed across three domains: Arterial inflow/forward flow, venous outflow/congestion, and microcirculatory perfusion. Suggested values are pragmatic reassessment triggers, not universal treatment thresholds. The aim is to identify whether ongoing hypoperfusion is driven

by reduced arterial inflow, venous congestion, combined inflow-congestion failure, or residual microcirculatory dysfunction. ABC: Airway, breathing, circulation; AVPG: Arterial-venous perfusion gradient; CO: Cardiac output; CI: Cardiac index; CRT: Capillary refill time; CVP: Central venous pressure; IAP: Intra-abdominal pressure; IO: Intraosseus; IV: Intravenous; IVC: Inferior vena cava; LVOT-VTI: Left ventricular outflow tract velocity-time integral; MAP: Mean arterial pressure; PE: Pulmonary embolism; PEEP: Positive end-expiratory pressure; PLR: Passive leg raise; POCUS: Point-of-care ultrasound; Pv-aCO_2: Veno-arterial carbon dioxide tension difference; RV: Right ventricle/right ventricular; $ScvO_2$: Central venous oxygen saturation; SVI: Stroke volume index; VExUS: Venous excess ultrasound score; TR: Tricuspid regurgitation.

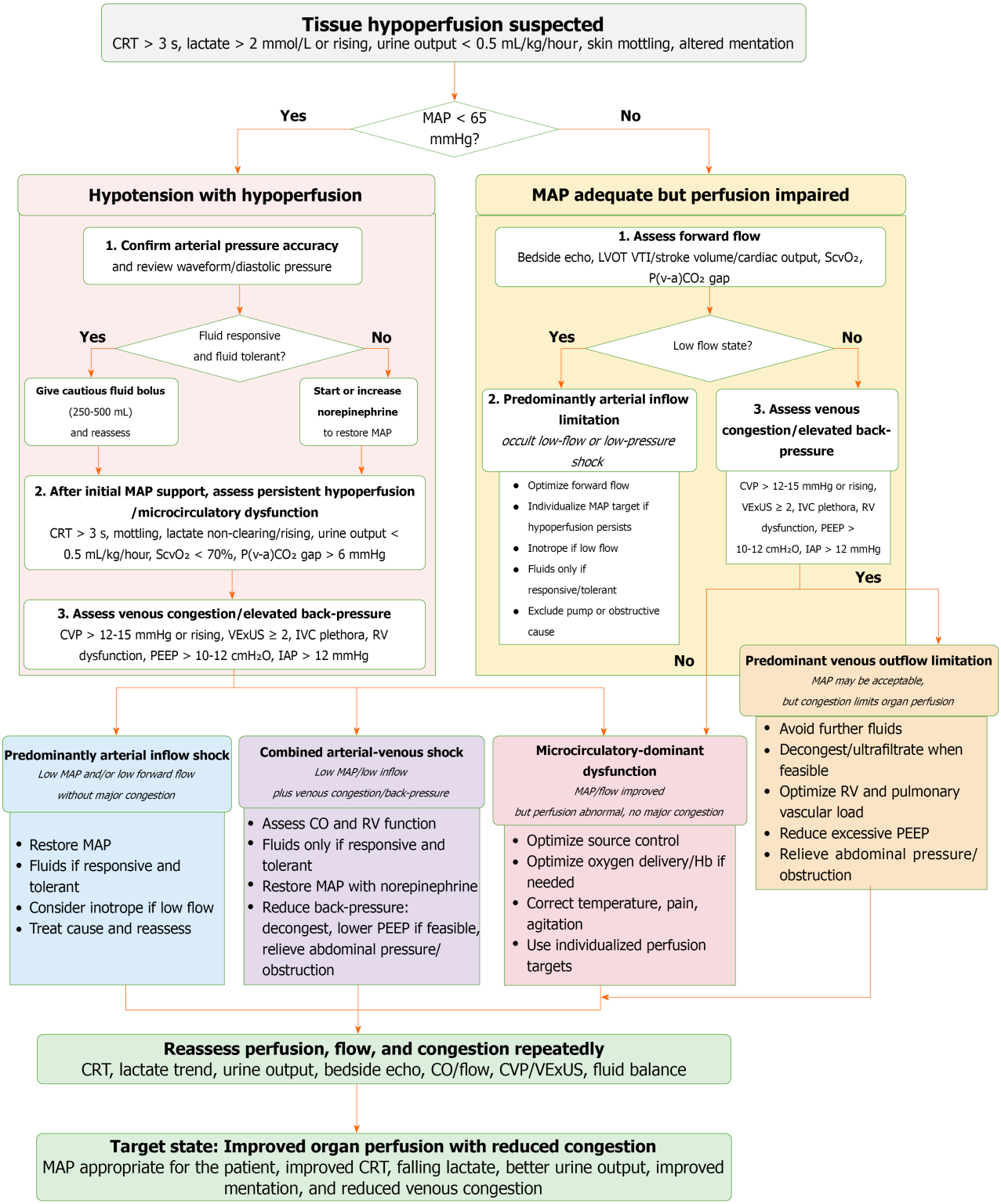


Figure 4 Arterial-venous perfusion gradient-guided bedside pathway for tissue hypoperfusion. The pathway integrates mean arterial pressure (MAP), forward flow, venous congestion, and tissue perfusion markers to classify suspected hypoperfusion into arterial inflow shock, predominant venous

congestion/venous outflow limitation, combined arterial-venous shock, or microcirculatory-dominant dysfunction. Suggested values such as MAP < 65 mmHg, capillary refill time > 3 seconds, lactate > 2 mmol/L or rising, urine output < 0.5 mL/kg/hour, fluid responsiveness, central venous pressure > 12-15 mmHg or rising, venous excess ultrasound score grade 2-3, right ventricle dysfunction, high positive end-expiratory pressure/dynamic hyperinflation, and intra-abdominal pressure > 12 mmHg are pragmatic reassessment triggers rather than universal treatment thresholds. Management is directed toward the dominant phenotype, followed by repeated reassessment of perfusion, flow, congestion, and organ function. The desired endpoint is improved organ perfusion with reduced venous congestion, rather than achievement of a single pressure target. AVPG: Arterial-venous perfusion gradient; CO: Cardiac output; CRT: Capillary refill time; CVP: Central venous pressure; Hb: Hemoglobin; IAP: Intra-abdominal pressure; IVC: Inferior vena cava; LVOT-VTI: Left ventricular outflow tract velocity-time integral; MAP: Mean arterial pressure; PEEP: Positive end-expiratory pressure; $P(v\text{-}a)CO_2$: Veno-arterial carbon dioxide tension difference; RV: Right ventricle/right ventricular; $ScvO_2$: Central venous oxygen saturation; VExUS: Venous excess ultrasound score.

FUTURE DIRECTIONS

Although growing physiological and clinical evidence highlights the role of venous congestion and microcirculatory dysfunction in critical illness, these dimensions are not yet consistently incorporated into routine hemodynamic decision-making. Current resuscitation strategies continue to rely predominantly on arterial pressure and cardiac output, while the influence of venous outflow pressure on tissue perfusion is less systematically addressed[1-3]. Further work is needed to define how AVPG assessment can be integrated into bedside practice and whether this approach improves clinically meaningful outcomes.

Prospective studies should evaluate whether resuscitation strategies guided by the AVPG improve organ function and clinical outcomes compared with conventional approaches based primarily on MAP or cardiac output. Such studies should incorporate simultaneous assessment of arterial inflow, venous congestion, and markers of tissue perfusion, and examine whether this integrated approach improves the alignment between macrocirculatory targets and tissue-level perfusion[11,41,57].

Ultrasound-based assessment of venous congestion represents a relevant area for further study. The role of indices such as the VExUS score in guiding fluid management and de-resuscitation strategies remains to be defined in prospective settings. Future studies should evaluate how these measures can be incorporated into structured resuscitation algorithms and combined with clinical and biochemical markers of perfusion. Another area of interest is integration of microcirculatory assessment into routine hemodynamic evaluation. Bedside indicators such as CRT, lactate kinetics, and the venous-to-arterial CO_2 difference are already available, but their use within structured decision-making frameworks remains variable. Clarifying how these markers can be combined with macrocirculatory and venous parameters to identify persistent perfusion abnormalities will be important.

Finally, stratification of patients according to the dominant physiological disturbance, whether reduced arterial inflow, venous congestion, or a combination of both, may allow more individualized resuscitation strategies. Prospective evaluation of such phenotype-based approaches may help determine whether tailoring therapy to the underlying mechanism of circulatory failure can improve organ function and clinical outcomes in critically ill patients[26,69].

CONCLUSION

Hemodynamic management in critical illness has traditionally prioritized arterial pressure and cardiac output. The evidence reviewed here suggests that elevated venous pressures and persistent microcirculatory dysfunction are also important drivers of organ hypoperfusion. Assessing venous congestion and tissue reperfusion markers alongside conventional arterial variables provides a more complete framework for evaluating circulatory failure at the bedside. The physiology-guided algorithm shows how arterial inflow, venous outflow pressure, and tissue perfusion markers can be woven into an iterative management approach. Future studies of these integrated perfusion-based strategies should help refine individualized resuscitation approaches that restore organ perfusion while limiting the harms of excessive fluid administration and unnecessary vasopressor escalation.

REFERENCES

1 **Vincent JL**, De Backer D. Circulatory shock. *N Engl J Med* 2013; **369**: 1726-1734 [*RCA*] [PMID: 24171518 DOI: 10.1056/NEJMra1208943] [FullText]

2 **Cecconi M**, De Backer D, Antonelli M, Beale R, Bakker J, Hofer C, Jaeschke R, Mebazaa A, Pinsky MR, Teboul JL, Vincent JL, Rhodes A. Consensus on circulatory shock and hemodynamic monitoring. Task force of the European Society of Intensive Care Medicine. *Intensive Care Med* 2014; **40**: 1795-1815 [*RCA*] [PMID: 25392034 DOI: 10.1007/s00134-014-3525-z] [FullText] [Full Text(PDF)]

3 **Singer M**, Deutschman CS, Seymour CW, Shankar-Hari M, Annane D, Bauer M, Bellomo R, Bernard GR, Chiche JD, Coopersmith CM, Hotchkiss RS, Levy MM, Marshall JC, Martin GS, Opal SM, Rubenfeld GD, van der Poll T, Vincent JL, Angus DC. The Third International Consensus Definitions for Sepsis and Septic Shock (Sepsis-3). *JAMA* 2016; **315**: 801-810 [*RCA*] [PMID: 26903338 DOI: 10.1001/jama.2016.0287] [FullText] [Full Text(PDF)]

4 **Evans L**, Rhodes A, Alhazzani W, Antonelli M, Coopersmith CM, French C, Machado FR, Mcintyre L, Ostermann M, Prescott HC, Schorr C, Simpson S, Wiersinga WJ, Alshamsi F, Angus DC, Arabi Y, Azevedo L, Beale R, Beilman G, Belley-Cote E, Burry L, Cecconi M, Centofanti J, Coz Yataco A, De Waele J, Dellinger RP, Doi K, Du B, Estenssoro E, Ferrer R, Gomersall C, Hodgson C, Møller MH, Iwashyna T, Jacob S,

Kleinpell R, Klompas M, Koh Y, Kumar A, Kwizera A, Lobo S, Masur H, McGloughlin S, Mehta S, Mehta Y, Mer M, Nunnally M, Oczkowski S, Osborn T, Papathanassoglou E, Perner A, Puskarich M, Roberts J, Schweickert W, Seckel M, Sevransky J, Sprung CL, Welte T, Zimmerman J, Levy M. Surviving sepsis campaign: international guidelines for management of sepsis and septic shock 2021. *Intensive Care Med* 2021; **47**: 1181-1247 [*RCA*] [PMID: 34599691 DOI: 10.1007/s00134-021-06506-y] [FullText] [Full Text(PDF)]

5 **Guyton AC**, Lindsey AW, Abernathy B, Richardson T. Venous return at various right atrial pressures and the normal venous return curve. *Am J Physiol* 1957; **189**: 609-615 [*RCA*] [PMID: 13458395 DOI: 10.1152/ajplegacy.1957.189.3.609] [FullText]

6 **Rothe CF**. Mean circulatory filling pressure: its meaning and measurement. *J Appl Physiol (1985)* 1993; **74**: 499-509 [*RCA*] [PMID: 8458763 DOI: 10.1152/jappl.1993.74.2.499] [FullText]

7 **Magder S**. Volume and its relationship to cardiac output and venous return. *Crit Care* 2016; **20**: 271 [*RCA*] [PMID: 27613307 DOI: 10.1186/s13054-016-1438-7] [FullText] [Full Text(PDF)]

8 **Berger D**, Takala J. Determinants of systemic venous return and the impact of positive pressure ventilation. *Ann Transl Med* 2018; **6**: 350 [*RCA*] [PMID: 30370277 DOI: 10.21037/atm.2018.05.27] [FullText]

9 **Ince C**. Hemodynamic coherence and the rationale for monitoring the microcirculation. *Crit Care* 2015; **19** Suppl 3: S8 [*RCA*] [PMID: 26729241 DOI: 10.1186/cc14726] [FullText] [Full Text(PDF)]

10 **Mullens W**, Abrahams Z, Francis GS, Sokos G, Taylor DO, Starling RC, Young JB, Tang WHW. Importance of venous congestion for worsening of renal function in advanced decompensated heart failure. *J Am Coll Cardiol* 2009; **53**: 589-596 [*RCA*] [PMID: 19215833 DOI: 10.1016/j.jacc.2008.05.068] [FullText] [Full Text(PDF)]

11 **Damman K**, Navis G, Smilde TD, Voors AA, van der Bij W, van Veldhuisen DJ, Hillege HL. Decreased cardiac output, venous congestion and the association with renal impairment in patients with cardiac dysfunction. *Eur J Heart Fail* 2007; **9**: 872-878 [*RCA*] [PMID: 17586090 DOI: 10.1016/j.ejheart.2007.05.010] [FullText]

12 **Legrand M**, Dupuis C, Simon C, Gayat E, Mateo J, Lukaszewicz AC, Payen D. Association between systemic hemodynamics and septic acute kidney injury in critically ill patients: a retrospective observational study. *Crit Care* 2013; **17**: R278 [*RCA*] [PMID: 24289206 DOI: 10.1186/cc13133] [FullText] [Full Text(PDF)]

13 **Chen CY**, Zhou Y, Wang P, Qi EY, Gu WJ. Elevated central venous pressure is associated with increased mortality and acute kidney injury in critically ill patients: a meta-analysis. *Crit Care* 2020; **24**: 80 [*RCA*] [PMID: 32138764 DOI: 10.1186/s13054-020-2770-5] [FullText] [Full Text(PDF)]

14 **Gelman S**. Venous function and central venous pressure: a physiologic story. *Anesthesiology* 2008; **108**: 735-748 [*RCA*] [PMID: 18362606 DOI: 10.1097/ALN.0b013e3181672607] [FullText]

15 **Magder S**. Bench-to-bedside review: An approach to hemodynamic monitoring--Guyton at the bedside. *Crit Care* 2012; **16**: 236 [*RCA*] [PMID: 23106914 DOI: 10.1186/cc11395] [FullText] [Full Text(PDF)]

16 **Vieillard-Baron A**, Matthay M, Teboul JL, Bein T, Schultz M, Magder S, Marini JJ. Experts' opinion on management of hemodynamics in ARDS patients: focus on the effects of mechanical ventilation. *Intensive Care Med* 2016; **42**: 739-749 [*RCA*] [PMID: 27038480 DOI: 10.1007/s00134-016-4326-3] [FullText]

17 **Giannakoulas G**, Farmakis IT, Hobohm L, Verbrugge FH, Tedford RJ, Sanz J. Acute right ventricular failure: pathophysiology, aetiology, assessment, and management. *Eur Heart J* 2025; **46**: 2520-2535 [*RCA*] [PMID: 40259766 DOI: 10.1093/eurheartj/ehaf215] [FullText]

18 **Lang RM**, Badano LP, Mor-Avi V, Afilalo J, Armstrong A, Ernande L, Flachskampf FA, Foster E, Goldstein SA, Kuznetsova T, Lancellotti P, Muraru D, Picard MH, Rietzschel ER, Rudski L, Spencer KT, Tsang W, Voigt JU. Recommendations for cardiac chamber quantification by echocardiography in adults: an update from the American Society of Echocardiography and the European Association of Cardiovascular Imaging. *J Am Soc Echocardiogr* 2015; **28**: 1-39.e14 [*RCA*] [PMID: 25559473 DOI: 10.1016/j.echo.2014.10.003] [FullText]

19 **De Backer D**, Ospina-Tascon G, Salgado D, Favory R, Creteur J, Vincent JL. Monitoring the microcirculation in the critically ill patient: current methods and future approaches. *Intensive Care Med* 2010; **36**: 1813-1825 [*RCA*] [PMID: 20689916 DOI: 10.1007/s00134-010-2005-3] [FullText]

20 **De Backer D**, Creteur J, Preiser JC, Dubois MJ, Vincent JL. Microvascular blood flow is altered in patients with sepsis. *Am J Respir Crit Care Med* 2002; **166**: 98-104 [*RCA*] [PMID: 12091178 DOI: 10.1164/rccm.200109-016oc] [FullText]

21 **Ince C**. The microcirculation is the motor of sepsis. *Crit Care* 2005; **9** Suppl 4: S13-S19 [*RCA*] [PMID: 16168069 DOI: 10.1186/cc3753] [Full Text] [Full Text(PDF)]

22 **Prowle JR**, Echeverri JE, Ligabo EV, Ronco C, Bellomo R. Fluid balance and acute kidney injury. *Nat Rev Nephrol* 2010; **6**: 107-115 [*RCA*] [PMID: 20027192 DOI: 10.1038/nrneph.2009.213] [FullText]

23 **Vellinga NA**, Ince C, Boerma EC. Elevated central venous pressure is associated with impairment of microcirculatory blood flow in sepsis: a hypothesis generating post hoc analysis. *BMC Anesthesiol* 2013; **13**: 17 [*RCA*] [PMID: 23919272 DOI: 10.1186/1471-2253-13-17] [FullText] [Full Text(PDF)]

24 **Pinsky MR**, García MIM, Dubin A. Significance of critical closing pressures (starling resistors) in arterial circulation. *Crit Care* 2024; **28**: 127 [*RCA*] [PMID: 38637877 DOI: 10.1186/s13054-024-04912-4] [FullText]

25 **Castro R**, Kattan E, Retamal J, Hernández G, Pinsky MR. Venous congestion from a vascular waterfall perspective: reframing congestion as a dynamic Starling resistor phenomenon. *Intensive Care Med Exp* 2025; **13**: 119 [*RCA*] [PMID: 41284176 DOI: 10.1186/s40635-025-00828-7] [FullText] [Full Text(PDF)]

26 **Huang L**, Huang Q, Ma W, Yang H. Understanding hemodynamic incoherence: Mechanisms, phenotypes, and implications for treatment. *Shock* 2025; **63**: 342-350 [*RCA*] [PMID: 39527481 DOI: 10.1097/SHK.0000000000002507] [FullText] [Full Text(PDF)]

27 **Muñoz F**, Born P, Bruna M, Ulloa R, González C, Philp V, Mondaca R, Blanco JP, Valenzuela ED, Retamal J, Miralles F, Wendel-Garcia PD, Ospina-Tascón GA, Castro R, Rola P, Bakker J, Hernández G, Kattan E. Coexistence of a fluid responsive state and venous congestion signals in critically ill patients: a multicenter observational proof-of-concept study. *Crit Care* 2024; **28**: 52 [*RCA*] [PMID: 38374167 DOI: 10.1186/s13054-024-04834-1] [FullText] [Full Text(PDF)]

28 **Boyd JH**, Forbes J, Nakada TA, Walley KR, Russell JA. Fluid resuscitation in septic shock: a positive fluid balance and elevated central venous pressure are associated with increased mortality. *Crit Care Med* 2011; **39**: 259-265 [*RCA*] [PMID: 20975548 DOI: 10.1097/CCM.0b013e3181feeb15] [FullText]

29 **Geri G**, Vignon P, Aubry A, Fedou AL, Charron C, Silva S, Repessé X, Vieillard-Baron A. Cardiovascular clusters in septic shock combining clinical and echocardiographic parameters: a post hoc analysis. *Intensive Care Med* 2019; **45**: 657-667 [*RCA*] [PMID: 30888443 DOI: 10.1007/s00134-019-05596-z] [FullText]

30 **Vieillard-Baron A**, Jardin F. Why protect the right ventricle in patients with acute respiratory distress syndrome? *Curr Opin Crit Care* 2003; **9**: 15-21 [*RCA*] [PMID: 12548024 DOI: 10.1097/00075198-200302000-00004] [FullText]

31 **Winton FR**. The influence of venous pressure on the isolated mammalian kidney. *J Physiol* 1931; **72**: 49-61 [*RCA*] [PMID: 16994199 DOI: 10.1113/jphysiol.1931.sp002761] [FullText] [Full Text(PDF)]

32 **Damman K**, van Deursen VM, Navis G, Voors AA, van Veldhuisen DJ, Hillege HL. Increased central venous pressure is associated with impaired renal function and mortality in a broad spectrum of patients with cardiovascular disease. *J Am Coll Cardiol* 2009; **53**: 582-588 [*RCA*] [PMID: 19215832 DOI: 10.1016/j.jacc.2008.08.080] [FullText]

33 **Panwar R**, McNicholas B, Teixeira JP, Kansal A. Renal perfusion pressure: role and implications in critical illness. *Ann Intensive Care* 2025; **15**: 115 [*RCA*] [PMID: 40775567 DOI: 10.1186/s13613-025-01535-y] [FullText] [Full Text(PDF)]

34 **Argaiz ER**, Rola P, Haycock KH, Verbrugge FH. Fluid management in acute kidney injury: from evaluating fluid responsiveness towards assessment of fluid tolerance. *Eur Heart J Acute Cardiovasc Care* 2022; **11**: 786-793 [*RCA*] [PMID: 36069621 DOI: 10.1093/ehjacc/zuac104] [FullText]

35 **Naschitz JE**, Slobodin G, Lewis RJ, Zuckerman E, Yeshurun D. Heart diseases affecting the liver and liver diseases affecting the heart. *Am Heart J* 2000; **140**: 111-120 [*RCA*] [PMID: 10874271 DOI: 10.1067/mhj.2000.107177] [FullText]

36 **Deitch EA**. Bacterial translocation or lymphatic drainage of toxic products from the gut: what is important in human beings? *Surgery* 2002; **131**: 241-244 [*RCA*] [PMID: 11894026 DOI: 10.1067/msy.2002.116408] [FullText]

37 **Beaubien-Souligny W**, Benkreira A, Robillard P, Bouabdallaoui N, Chassé M, Desjardins G, Lamarche Y, White M, Bouchard J, Denault A. Alterations in Portal Vein Flow and Intrarenal Venous Flow Are Associated With Acute Kidney Injury After Cardiac Surgery: A Prospective Observational Cohort Study. *J Am Heart Assoc* 2018; **7**: e009961 [*RCA*] [PMID: 30371304 DOI: 10.1161/JAHA.118.009961] [FullText]

38 **Soliman Aboumarie H**, Nasa P, Malbrain MLNG. Cardiovascular effects of intra-abdominal hypertension: current perspectives. *Anaesthesiol Intensive Ther* 2025; **57**: 340-355 [*RCA*] [PMID: 41362929 DOI: 10.5114/ait/210612] [FullText] [Full Text(PDF)]

39 **Unnerbäck M**, Ottesen JT, Reinstrup P. Increased Intracranial Pressure Attenuates the Pulsating Component of Cerebral Venous Outflow. *Neurocrit Care* 2019; **31**: 273-279 [*RCA*] [PMID: 31240621 DOI: 10.1007/s12028-019-00733-4] [FullText] [Full Text(PDF)]

40 **Silva PL**, Ball L, Rocco PRM, Pelosi P. Physiological and Pathophysiological Consequences of Mechanical Ventilation. *Semin Respir Crit Care Med* 2022; **43**: 321-334 [*RCA*] [PMID: 35439832 DOI: 10.1055/s-0042-1744447] [FullText]

41 **Pinsky MR**. Cardiovascular issues in respiratory care. *Chest* 2005; **128**: 592S-597S [*RCA*] [PMID: 16306058 DOI: 10.1378/chest.128.5_suppl_2.592S] [FullText]

42 **Malbrain ML**, Chiumello D, Pelosi P, Bihari D, Innes R, Ranieri VM, Del Turco M, Wilmer A, Brienza N, Malcangi V, Cohen J, Japiassu A, De Keulenaer BL, Daelemans R, Jacquet L, Laterre PF, Frank G, de Souza P, Cesana B, Gattinoni L. Incidence and prognosis of intraabdominal hypertension in a mixed population of critically ill patients: a multiple-center epidemiological study. *Crit Care Med* 2005; **33**: 315-322 [*RCA*] [PMID: 15699833 DOI: 10.1097/01.ccm.0000153408.09806.1b] [FullText]

43 **Malbrain ML**, Marik PE, Witters I, Cordemans C, Kirkpatrick AW, Roberts DJ, Van Regenmortel N. Fluid overload, de-resuscitation, and outcomes in critically ill or injured patients: a systematic review with suggestions for clinical practice. *Anaesthesiol Intensive Ther* 2014; **46**: 361-380 [*RCA*] [PMID: 25432556 DOI: 10.5603/AIT.2014.0060] [FullText]

44 **Monnet X**, Messina A, Greco M, Bakker J, Aissaoui N, Cecconi M, Coppalini G, De Backer D, Edul VK, Evans L, Hernández G, Hunsicker O, Ince C, Kaufmann T, Levy B, Malbrain MLNG, Mebazaa A, Myatra SN, Ostermann M, Pinsky MR, Saugel B, Savi M, Singer M, Teboul JL, Vieillard-Baron A, Vincent JL, Chew MS. ESICM guidelines on circulatory shock and hemodynamic monitoring 2025. *Intensive Care Med* 2025; **51**: 1971-2012 [*RCA*] [PMID: 41236566 DOI: 10.1007/s00134-025-08137-z] [FullText]

45 **ANDROMEDA-SHOCK-2 Investigators for the ANDROMEDA Research Network**, Spanish Society of Anesthesiology, Reanimation and Pain Therapy (SEDAR), and Latin American Intensive Care Network (LIVEN), Hernandez G, Ospina-Tascón GA, Kattan E, Ibarra-Estrada M, Ramasco F, Orozco N, Ramos K, Aldana JL, Ferri G, Hamzaoui O, De Backer D, Teboul JL, Vieillard-Baron A, Petri Damiani L, García-Gallardo GA, Morales S, Carmona Garcia P, Mendez R, Hernandez-Gilsoul T, Pérez-Nieto OR, Olea Vielba C, Ramos S, Dominguez D, Bruna M, David S, Wendel-Garcia PD, Galiana-Ivars M, Myatra SN, Messina A, Cecconi M, Pozo M, Amthauer M, Higuera E, Al Duhalib Z, Rico-Feijoo J, Ferrer-Gómez C, Pérez-Carbonell A, Martinez-Castro S, Redondo Calvo FJ, Vives M, Sanchez HF, Bilbao I, Fernandez P, Al-Fares A, Benitez-Cano A, Gonzalez C, Reyes LF, Rodriguez-Guillen JH, Cristino AV, Pendino JC, Ortiz G, Alonso-Gonzalez MC, Murias G, Aguirre-Ávalos G, Hernández L, Calderón Barajas ZA, Zarragoikoetxea I, Monnet X, Goury A, Mendonça Dos Santos T, Vallecilla L, Martins de Lima L, Sady E, Alegria L, Ostermann M, Bakker J, Biasi Cavalcanti A. Personalized Hemodynamic Resuscitation Targeting Capillary Refill Time in Early Septic Shock: The ANDROMEDA-SHOCK-2 Randomized Clinical Trial. *JAMA* 2025; **334**: 1988-1999 [*RCA*] [PMID: 41159835 DOI: 10.1001/jama.2025.20402] [FullText] [Full Text(PDF)]

46 **Jansen TC**, van Bommel J, Schoonderbeek FJ, Sleeswijk Visser SJ, van der Klooster JM, Lima AP, Willemsen SP, Bakker J; LACTATE study group. Early lactate-guided therapy in intensive care unit patients: a multicenter, open-label, randomized controlled trial. *Am J Respir Crit Care Med* 2010; **182**: 752-761 [*RCA*] [PMID: 20463176 DOI: 10.1164/rccm.200912-1918OC] [FullText]

47 **Ltaief Z**, Schneider AG, Liaudet L. Pathophysiology and clinical implications of the veno-arterial PCO(2) gap. *Crit Care* 2021; **25**: 318 [*RCA*] [PMID: 34461974 DOI: 10.1186/s13054-021-03671-w] [FullText]

48 **Magder S**. The meaning of blood pressure. *Crit Care* 2018; **22**: 257 [*RCA*] [PMID: 30305136 DOI: 10.1186/s13054-018-2171-1] [FullText] [Full Text(PDF)]

49 **De Backer D**, Orbegozo Cortes D, Donadello K, Vincent JL. Pathophysiology of microcirculatory dysfunction and the pathogenesis of septic shock. *Virulence* 2014; **5**: 73-79 [*RCA*] [PMID: 24067428 DOI: 10.4161/viru.26482] [FullText] [Full Text(PDF)]

50 **Sakr Y**, Dubois MJ, De Backer D, Creteur J, Vincent JL. Persistent microcirculatory alterations are associated with organ failure and death in patients with septic shock. *Crit Care Med* 2004; **32**: 1825-1831 [*RCA*] [PMID: 15343008 DOI: 10.1097/01.ccm.0000138558.16257.3f] [Full Text]

51 **Bruno RR**, Wollborn J, Fengler K, Flick M, Wunder C, Allgäuer S, Thiele H, Schemmelmann M, Hornemann J, Moecke HME, Demirtas F, Palici L, Franz M, Saugel B, Kattan E, De Backer D, Bakker J, Hernandez G, Kelm M, Jung C. Direct assessment of microcirculation in shock: a randomized-controlled multicenter study. *Intensive Care Med* 2023; **49**: 645-655 [*RCA*] [PMID: 37278760 DOI: 10.1007/s00134-023-07098-5] [FullText] [Full Text(PDF)]

52 **Yoshida T**, Yoshida T, Noma H, Nomura T, Suzuki A, Mihara T. Diagnostic accuracy of point-of-care ultrasound for shock: a systematic review and meta-analysis. *Crit Care* 2023; **27**: 200 [*RCA*] [PMID: 37231510 DOI: 10.1186/s13054-023-04495-6] [FullText] [Full Text(PDF)]

53 **Orso D**, Paoli I, Piani T, Cilenti FL, Cristiani L, Guglielmo N. Accuracy of Ultrasonographic Measurements of Inferior Vena Cava to

Determine Fluid Responsiveness: A Systematic Review and Meta-Analysis. *J Intensive Care Med* 2020; **35**: 354-363 [*RCA*] [PMID: 29343170 DOI: 10.1177/0885066617752308] [FullText]

54 **Beaubien-Souligny W**, Rola P, Haycock K, Bouchard J, Lamarche Y, Spiegel R, Denault AY. Quantifying systemic congestion with Point-Of-Care ultrasound: development of the venous excess ultrasound grading system. *Ultrasound J* 2020; **12**: 16 [*RCA*] [PMID: 32270297 DOI: 10.1186/s13089-020-00163-w] [FullText] [Full Text(PDF)]

55 **Longino A**, Martin K, Leyba K, Siegel G, Thai TN, Riscinti M, Douglas IS, Gill E, Burke J. Prospective Evaluation of Venous Excess Ultrasound for Estimation of Venous Congestion. *Chest* 2024; **165**: 590-600 [*RCA*] [PMID: 37813180 DOI: 10.1016/j.chest.2023.09.029] [Full Text] [Full Text(PDF)]

56 **Melo RH**, Gioli-Pereira L, Melo E, Rola P. Venous excess ultrasound score association with acute kidney injury in critically ill patients: a systematic review and meta-analysis of observational studies. *Ultrasound J* 2025; **17**: 16 [*RCA*] [PMID: 40029471 DOI: 10.1186/s13089-025-00413-9] [FullText]

57 **Andrei S**, Bahr PA, Nguyen M, Bouhemad B, Guinot PG. Prevalence of systemic venous congestion assessed by Venous Excess Ultrasound Grading System (VExUS) and association with acute kidney injury in a general ICU cohort: a prospective multicentric study. *Crit Care* 2023; **27**: 224 [*RCA*] [PMID: 37291662 DOI: 10.1186/s13054-023-04524-4] [FullText]

58 **Hernández G**, Ospina-Tascón GA, Damiani LP, Estenssoro E, Dubin A, Hurtado J, Friedman G, Castro R, Alegría L, Teboul JL, Cecconi M, Ferri G, Jibaja M, Pairumani R, Fernández P, Barahona D, Granda-Luna V, Cavalcanti AB, Bakker J; The ANDROMEDA SHOCK Investigators and the Latin America Intensive Care Network (LIVEN), Hernández G, Ospina-Tascón G, Petri Damiani L, Estenssoro E, Dubin A, Hurtado J, Friedman G, Castro R, Alegría L, Teboul JL, Cecconi M, Cecconi M, Ferri G, Jibaja M, Pairumani R, Fernández P, Barahona D, Cavalcanti AB, Bakker J, Hernández G, Alegría L, Ferri G, Rodriguez N, Holger P, Soto N, Pozo M, Bakker J, Cook D, Vincent JL, Rhodes A, Kavanagh BP, Dellinger P, Rietdijk W, Carpio D, Pavéz N, Henriquez E, Bravo S, Valenzuela ED, Vera M, Dreyse J, Oviedo V, Cid MA, Larroulet M, Petruska E, Sarabia C, Gallardo D, Sanchez JE, González H, Arancibia JM, Muñoz A, Ramirez G, Aravena F, Aquevedo A, Zambrano F, Bozinovic M, Valle F, Ramirez M, Rossel V, Muñoz P, Ceballos C, Esveile C, Carmona C, Candia E, Mendoza D, Sanchez A, Ponce D, Ponce D, Lastra J, Nahuelpán B, Fasce F, Luengo C, Medel N, Cortés C, Campassi L, Rubatto P, Horna N, Furche M, Pendino JC, Bettini L, Lovesio C, González MC, Rodruguez J, Canales H, Caminos F, Galletti C, Minoldo E, Aramburu MJ, Olmos D, Nin N, Tenzi J, Quiroga C, Lacuesta P, Gaudín A, Pais R, Silvestre A, Olivera G, Rieppi G, Berrutti D, Ochoa M, Cobos P, Vintimilla F, Ramirez V, Tobar M, García F, Picoita F, Remache N, Granda V, Paredes F, Barzallo E, Garcés P, Guerrero F, Salazar S, Torres G, Tana C, Calahorrano J, Solis F, Torres P, Herrera L, Ornes A, Peréz V, Delgado G, López A, Espinosa E, Moreira J, Salcedo B, Villacres I, Suing J, Lopez M, Gomez L, Toctaquiza G, Cadena Zapata M, Orazabal MA, Pardo Espejo R, Jimenez J, Calderón A, Paredes G, Barberán JL, Moya T, Atehortua H, Sabogal R, Ortiz G, Lara A, Sanchez F, Hernán Portilla A, Dávila H, Mora JA, Calderón LE, Alvarez I, Escobar E, Bejarano A, Bustamante LA, Aldana JL. Effect of a Resuscitation Strategy Targeting Peripheral Perfusion Status vs Serum Lactate Levels on 28-Day Mortality Among Patients With Septic Shock: The ANDROMEDA-SHOCK Randomized Clinical Trial. *JAMA* 2019; **321**: 654-664 [*RCA*] [PMID: 30772908 DOI: 10.1001/jama.2019.0071] [FullText] [Full Text(PDF)]

59 **Monnet X**, Shi R, Teboul JL. Prediction of fluid responsiveness. What's new? *Ann Intensive Care* 2022; **12**: 46 [*RCA*] [PMID: 35633423 DOI: 10.1186/s13613-022-01022-8] [FullText] [Full Text(PDF)]

60 **Silversides JA**, Major E, Ferguson AJ, Mann EE, McAuley DF, Marshall JC, Blackwood B, Fan E. Conservative fluid management or deresuscitation for patients with sepsis or acute respiratory distress syndrome following the resuscitation phase of critical illness: a systematic review and meta-analysis. *Intensive Care Med* 2017; **43**: 155-170 [*RCA*] [PMID: 27734109 DOI: 10.1007/s00134-016-4573-3] [FullText]

61 **Malbrain MLNG**, Martin G, Ostermann M. Everything you need to know about deresuscitation. *Intensive Care Med* 2022; **48**: 1781-1786 [*RCA*] [PMID: 35932335 DOI: 10.1007/s00134-022-06761-7] [FullText] [Full Text(PDF)]

62 **Meyhoff TS**, Hjortrup PB, Wetterslev J, Sivapalan P, Laake JH, Cronhjort M, Jakob SM, Cecconi M, Nalos M, Ostermann M, Malbrain M, Pettilä V, Møller MH, Kjær MN, Lange T, Overgaard-Steensen C, Brand BA, Winther-Olesen M, White JO, Quist L, Westergaard B, Jonsson AB, Hjortsø CJS, Meier N, Jensen TS, Engstrøm J, Nebrich L, Andersen-Ranberg NC, Jensen JV, Joseph NA, Poulsen LM, Herløv LS, Sølling CG, Pedersen SK, Knudsen KK, Straarup TS, Vang ML, Bundgaard H, Rasmussen BS, Aagaard SR, Hildebrandt T, Russell L, Bestle MH, Schønemann-Lund M, Brøchner AC, Elvander CF, Hoffmann SKL, Rasmussen ML, Martin YK, Friberg FF, Seter H, Aslam TN, Ådnøy S, Seidel P, Strand K, Johnstad B, Joelsson-Alm E, Christensen J, Ahlstedt C, Pfortmueller CA, Siegemund M, Greco M, Raděj J, Kříž M, Gould DW, Rowan KM, Mouncey PR, Perner A; CLASSIC Trial Group. Restriction of Intravenous Fluid in ICU Patients with Septic Shock. *N Engl J Med* 2022; **386**: 2459-2470 [*RCA*] [PMID: 35709019 DOI: 10.1056/NEJMoa2202707] [FullText]

63 **National Heart**, Lung, and Blood Institute Prevention and Early Treatment of Acute Lung Injury Clinical Trials Network, Shapiro NI, Douglas IS, Brower RG, Brown SM, Exline MC, Ginde AA, Gong MN, Grissom CK, Hayden D, Hough CL, Huang W, Iwashyna TJ, Jones AE, Khan A, Lai P, Liu KD, Miller CD, Oldmixon K, Park PK, Rice TW, Ringwood N, Semler MW, Steingrub JS, Talmor D, Thompson BT, Yealy DM, Self WH. Early Restrictive or Liberal Fluid Management for Sepsis-Induced Hypotension. *N Engl J Med* 2023; **388**: 499-510 [*RCA*] [PMID: 36688507 DOI: 10.1056/NEJMoa2212663] [FullText] [Full Text(PDF)]

64 **Bakker J**, Kattan E, Annane D, Castro R, Cecconi M, De Backer D, Dubin A, Evans L, Gong MN, Hamzaoui O, Ince C, Levy B, Monnet X, Ospina Tascón GA, Ostermann M, Pinsky MR, Russell JA, Saugel B, Scheeren TWL, Teboul JL, Vieillard Baron A, Vincent JL, Zampieri FG, Hernandez G. Current practice and evolving concepts in septic shock resuscitation. *Intensive Care Med* 2022; **48**: 148-163 [*RCA*] [PMID: 34910228 DOI: 10.1007/s00134-021-06595-9] [FullText]

65 **Rola P**, Miralles-Aguiar F, Argaiz E, Beaubien-Souligny W, Haycock K, Karimov T, Dinh VA, Spiegel R. Clinical applications of the venous excess ultrasound (VExUS) score: conceptual review and case series. *Ultrasound J* 2021; **13**: 32 [*RCA*] [PMID: 34146184 DOI: 10.1186/s13089-021-00232-8] [FullText] [Full Text(PDF)]

66 **McGuire WC**, Sullivan L, Odish MF, Desai B, Morris TA, Fernandes TM. Management Strategies for Acute Pulmonary Embolism in the ICU. *Chest* 2024; **166**: 1532-1545 [*RCA*] [PMID: 38830402 DOI: 10.1016/j.chest.2024.04.032] [FullText]

67 **Thiele H**, Zeymer U, Akin I, Behnes M, Rassaf T, Mahabadi AA, Lehmann R, Eitel I, Graf T, Seidler T, Schuster A, Skurk C, Duerschmied D, Clemmensen P, Hennersdorf M, Fichtlscherer S, Voigt I, Seyfarth M, John S, Ewen S, Linke A, Tigges E, Nordbeck P, Bruch L, Jung C, Franz J, Lauten P, Goslar T, Feistritzer HJ, Pöss J, Kirchhof E, Ouarrak T, Schneider S, Desch S, Freund A; ECLS-SHOCK Investigators. Extracorporeal Life Support in Infarct-Related Cardiogenic Shock. *N Engl J Med* 2023; **389**: 1286-1297 [*RCA*] [PMID: 37634145 DOI: 10.1056/NEJMoa2307227] [FullText]

68 **Møller JE**, Engstrøm T, Jensen LO, Eiskjær H, Mangner N, Polzin A, Schulze PC, Skurk C, Nordbeck P, Clemmensen P, Panoulas V, Zimmer S, Schäfer A, Werner N, Frydland M, Holmvang L, Kjærgaard J, Sørensen R, Lønborg J, Lindholm MG, Udesen NLJ, Junker A, Schmidt H, Terkelsen CJ, Christensen S, Christiansen EH, Linke A, Woitek FJ, Westenfeld R, Möbius-Winkler S, Wachtell K, Ravn HB, Lassen JF,

Boesgaard S, Gerke O, Hassager C; DanGer Shock Investigators. Microaxial Flow Pump or Standard Care in Infarct-Related Cardiogenic Shock. *N Engl J Med* 2024; **390**: 1382-1393 [*RCA*] [PMID: 38587239 DOI: 10.1056/NEJMoa2312572] [FullText]

69 **Ezad SM**, Ryan M, Donker DW, Pappalardo F, Barrett N, Camporota L, Price S, Kapur NK, Perera D. Unloading the Left Ventricle in Venoarterial ECMO: In Whom, When, and How? *Circulation* 2023; **147**: 1237-1250 [*RCA*] [PMID: 37068133 DOI: 10.1161/CIRCULATIONAHA.122.062371] [FullText]

70 **Seymour CW**, Kennedy JN, Wang S, Chang CH, Elliott CF, Xu Z, Berry S, Clermont G, Cooper G, Gomez H, Huang DT, Kellum JA, Mi Q, Opal SM, Talisa V, van der Poll T, Visweswaran S, Vodovotz Y, Weiss JC, Yealy DM, Yende S, Angus DC. Derivation, Validation, and Potential Treatment Implications of Novel Clinical Phenotypes for Sepsis. *JAMA* 2019; **321**: 2003-2017 [*RCA*] [PMID: 31104070 DOI: 10.1001/jama.2019.5791] [FullText] [Full Text(PDF)]

71 **Saura O**, Hékimian G, Del Marmol G, Lucenteforte M, Pineton de Chambrun M, Chommeloux J, Assouline B, Petit M, Juvin C, Gautier M, Moyon Q, Visinoni N, Luyt CE, Leprince P, Lebreton G, Schmidt M, Combes A, Levy D; PAPO-Flow Study Investigators. Effect of ECMO Flow Variations on Pulmonary Capillary Wedge Pressure in Patients With Cardiogenic Shock. *J Am Coll Cardiol* 2025; **86**: 768-778 [*RCA*] [PMID: 40930613 DOI: 10.1016/j.jacc.2025.06.048] [FullText]

FOOTNOTES

Specialty type: Critical care medicine

Country of origin: India

Author contributions: Kataria S conceived and conceptualized the study, performed the primary literature review, drafted the manuscript, prepared all images; Vinjamuri S and Goel S contributed to literature review and manuscript refinement; Juneja D contributed to conceptualization, critical intellectual input, substantial manuscript revision, and overall supervision of the work; and all authors approved the final manuscript. Kataria S conceived and conceptualized the study, conducted the primary literature review, drafted the manuscript, and prepared all images. Juneja D contributed to conceptualization, critical intellectual input, substantial manuscript revision, and overall supervision of the work. Both Kataria S and Juneja D have played crucial and indispensable roles in conceptualizing the manuscript, literature synthesis and interpretation, and manuscript preparation. They have made vital contributions towards the completion of the project and thus qualify as the co-corresponding authors.

AI contribution statement: The authors used Grammarly AI tool for language polishing, grammar correction, and improvement of readability during manuscript preparation.

Conflict-of-interest statement: All authors declare that they have no conflict of interest to disclose.

S-Editor: Liu JH
L-Editor: A
P-Editor: Wang CH

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